Machine Learning

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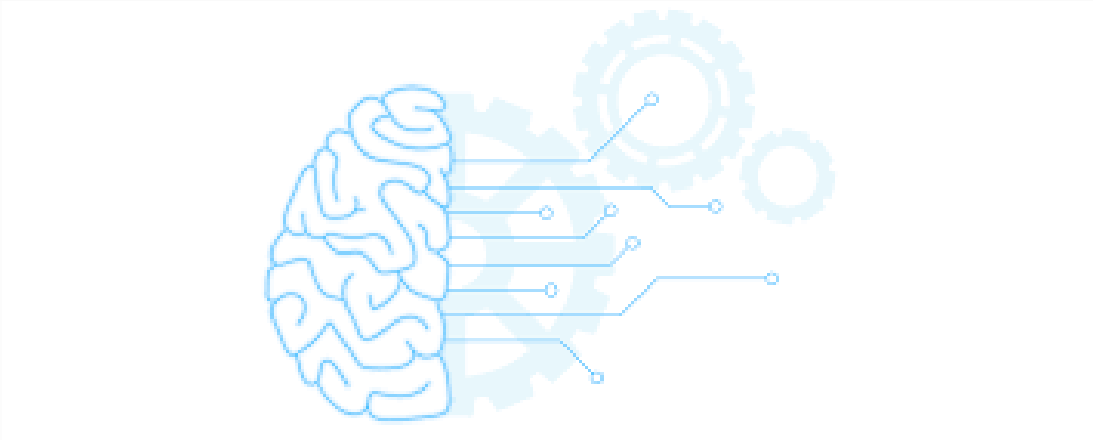
What is Machine Learning?

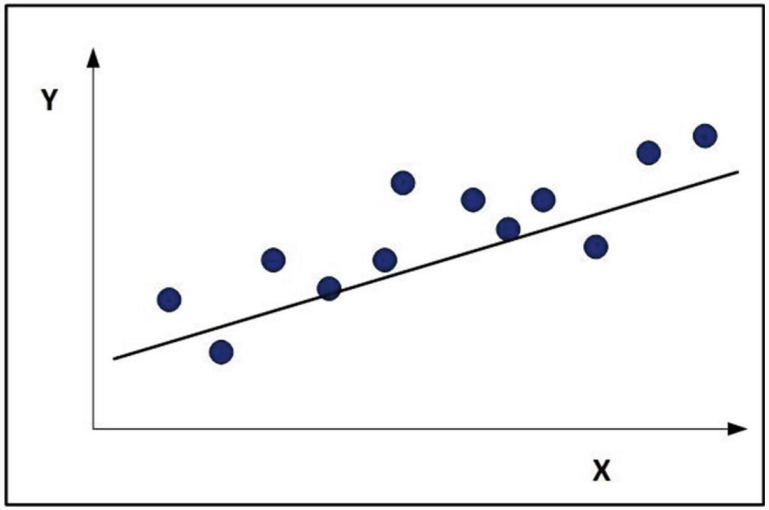
Machine Learning : the field of study that gives computers the ability to learn without being explicitly programmed.

- Supervised Learning
- Unsupervised Learning
- Reinforcement learning

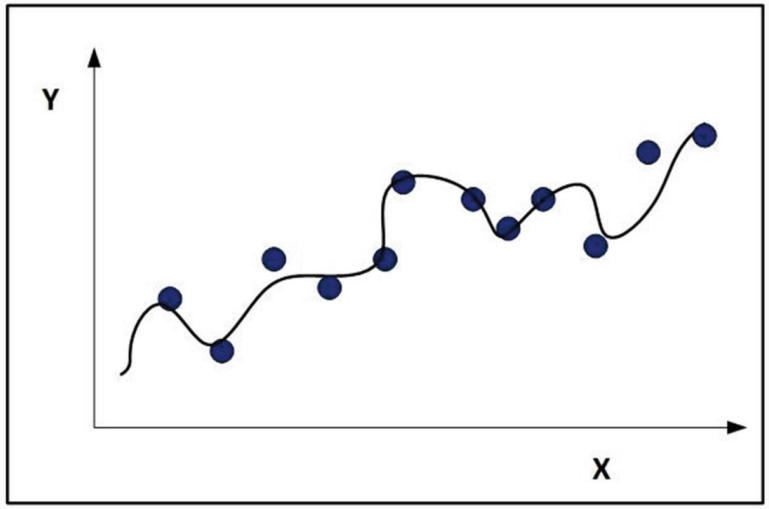


Overfitting, Underfitting

We want to construct a prediction model for the followings:



Underfitting



Overfitting

Bias, Variance

Bias(편향) : the expected error created by using a model to approximate a real world.

$$\text{Bias}\left[\hat{f}(x)\right] = \mathbb{E}\left[\hat{f}(x) - f(x)\right]$$

$$\text{High Bias} \iff \text{Underfitting}$$

Variance(분산) : the amount our predicted values would change if we had a different training dataset.

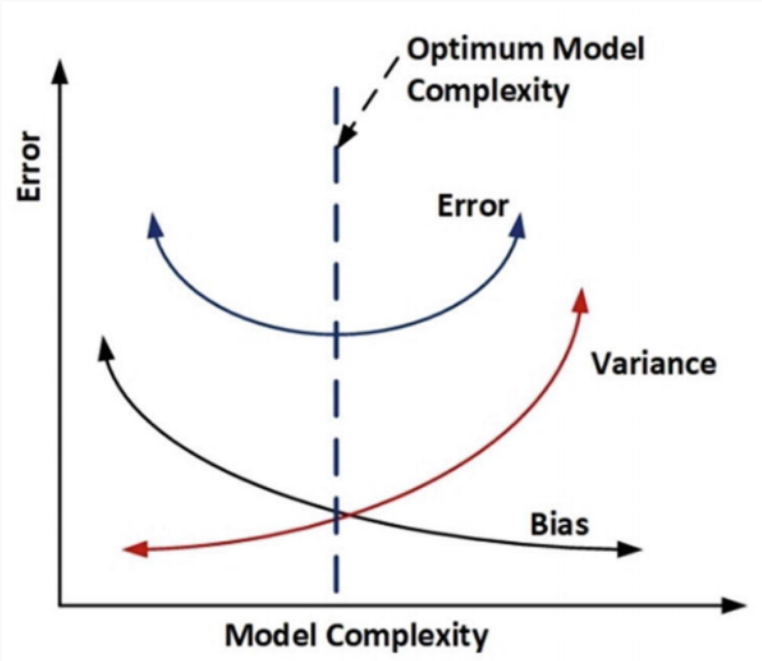
$$\text{Var}\left[\hat{f}(x)\right] = \mathbb{E}\left[\hat{f}(x)^2\right] - \mathbb{E}\left[\hat{f}(x)\right]^2$$

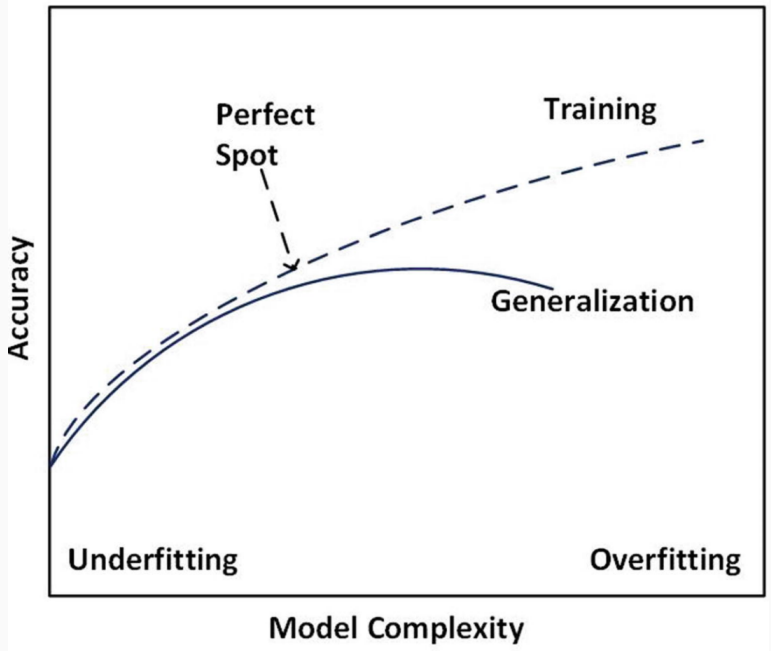
$$\text{High Variance} \iff \text{Overfitting}$$

f : objective function, $\hat{f}$: prediction function.

Bias-Variance Tradeoff

: the property of a model that the variance of the parameter estimated across samples can be reduced by increasing the bias in the estimated parameters.





Accuracy, Regularization

- Accuracy

$$\text{accuracy} = \frac{\text{number of correctly classified cases}}{\text{total number of cases}}$$

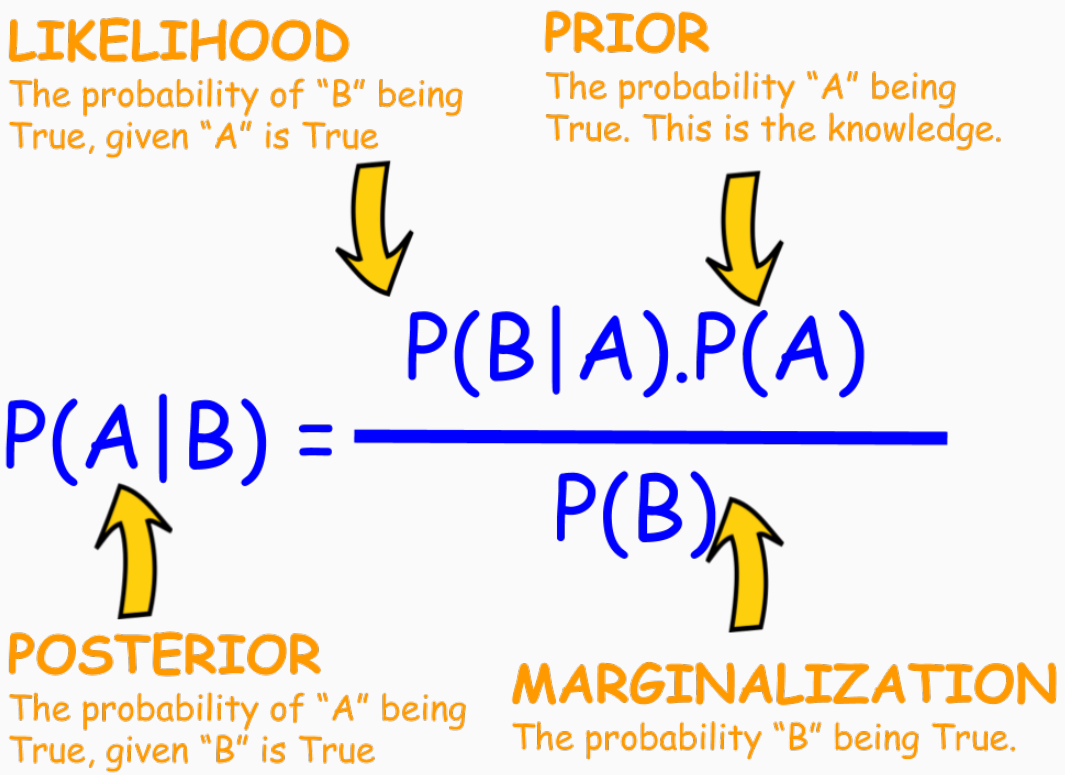
$$\text{error} = 1 - \text{accuracy}$$

- Regularization

$$C(\tau) = \text{regularizer} + \text{loss}(\tau) = \alpha(\lambda) + \text{loss}(\tau),$$

C: cost function, $\alpha \geq 0$: regularizer or penalty function

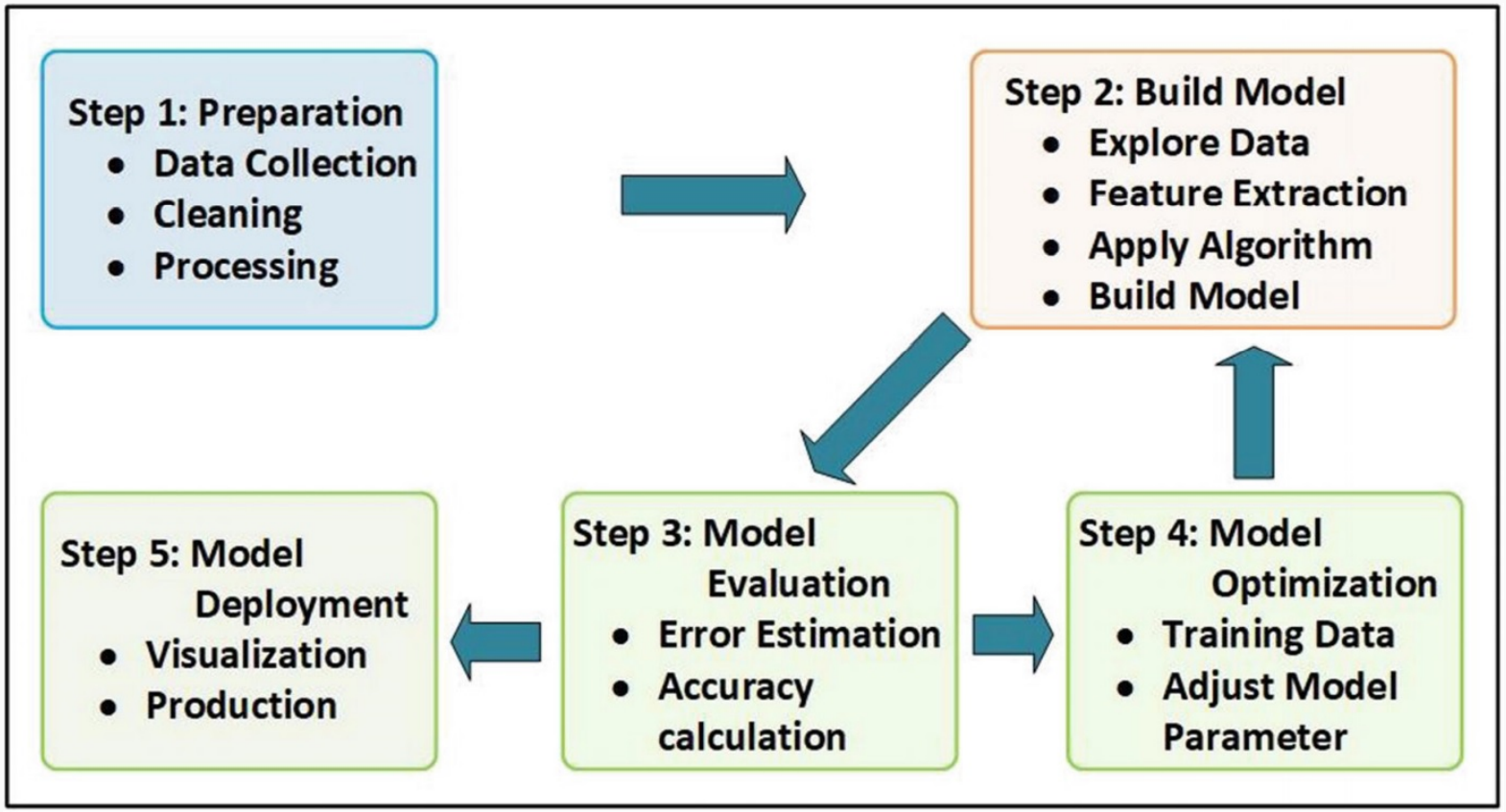
Bayesian Learning



A: hypothesis whose probability may be affected by 'evidence',

B: evidence.

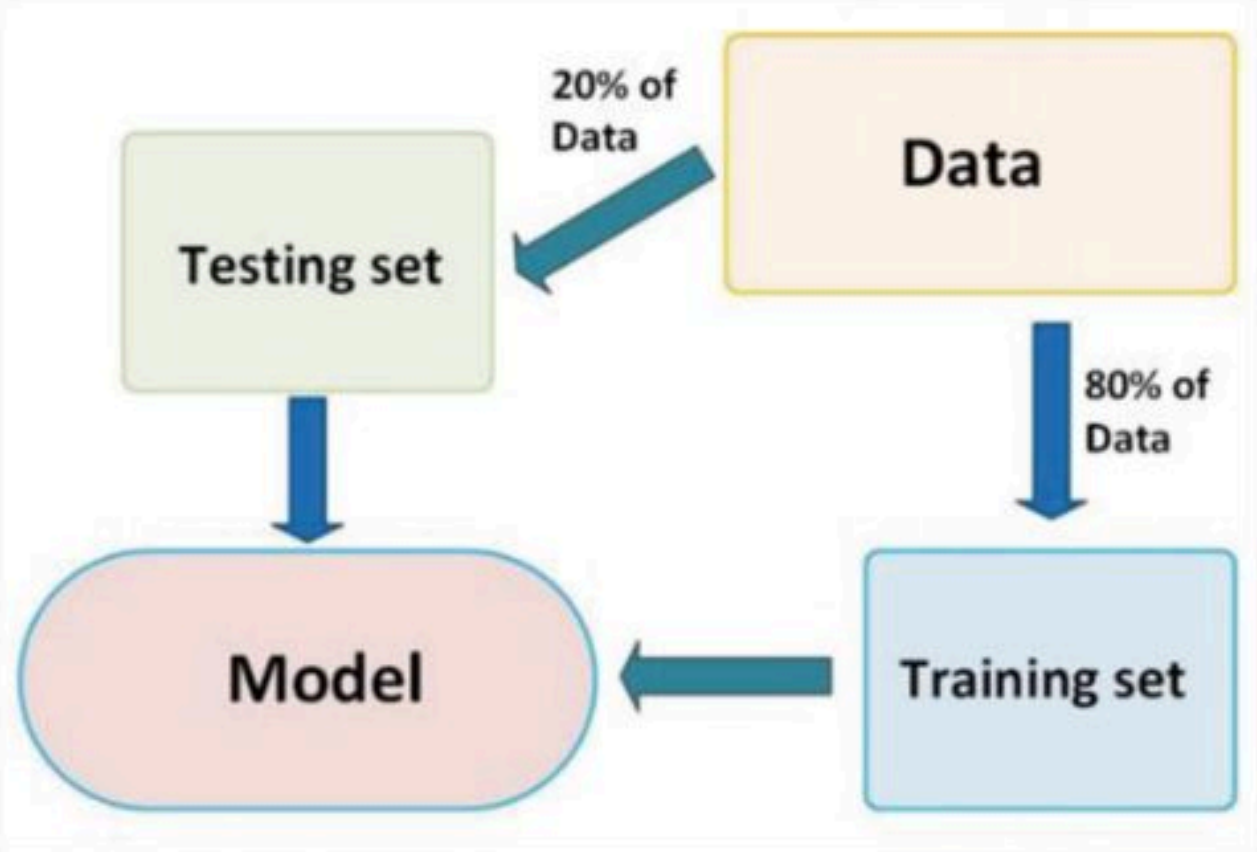
Machine Learning Workflow



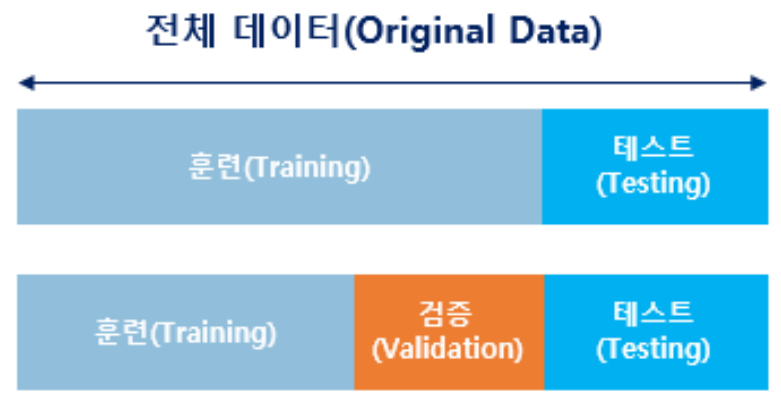
Machine Learning Workflow

Validation of Models

The Hold-Out Method



Validation of Models: The Cross-Validation Method



Note) Testing set is not used in learning. Only once used when we measure the ability.

- k-fold cross-validation

Runs	Splits						Accuracy
1	Test	Train	Train	Train	Train	Train	80
2	Train	Test	Train	Train	Train	Train	82
3	Train	Train	Test	Train	Train	Train	88
4	Train	Train	Train	Test	Train	Train	90
5	Train	Train	Train	Train	Test	Train	85
6	Train	Train	Train	Train	Train	Test	92

6-fold cross validation

Example(Python): k-fold cross validation

```python
# Import Libraries
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn import svm # import (SVM) for classification
from sklearn import datasets
iris = datasets.load_iris() # load the data
# Data split into train/test sets with 25% for testing
X_train, X_test, y_train, y_test = train_test_split(iris.data, iris.target, test_size=0.25,
                                                    random_state=0)
# Train data for Linear Support Vector Classifier (SVC) model for prediction
clf = svm.SVC(kernel='linear', C=1).fit(X_train, y_train)
# Performance measurement with the test data
clf.score(X_test, y_test)
```

✓ 0.1s

```
0.9736842105263158
```

Example(Python): k-fold cross validation

```python
svc = svm.SVC(kernel='linear', C=1).fit(X_train, y_train)
# cross_val_score a model, the data set, and the number of folds:
cvs = cross_val_score(svc, iris.data, iris.target, cv=5)
# Print accuracy of each fold:
print(cvs)
# Mean accuracy of all 5 folds:
print(cvs.mean())
```

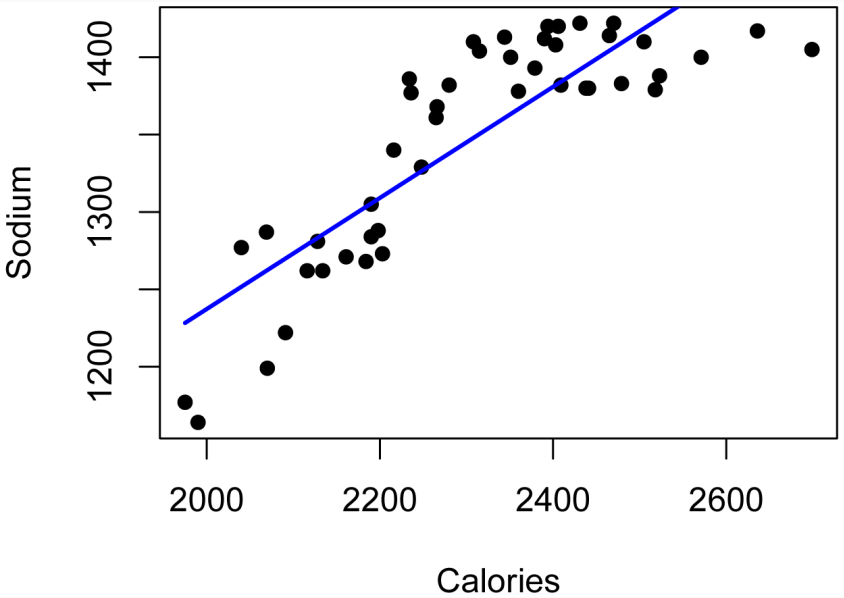
✓ 0.3s

```
[0.96666667 1.         0.96666667 0.96666667 1.        ]
0.9800000000000001
```

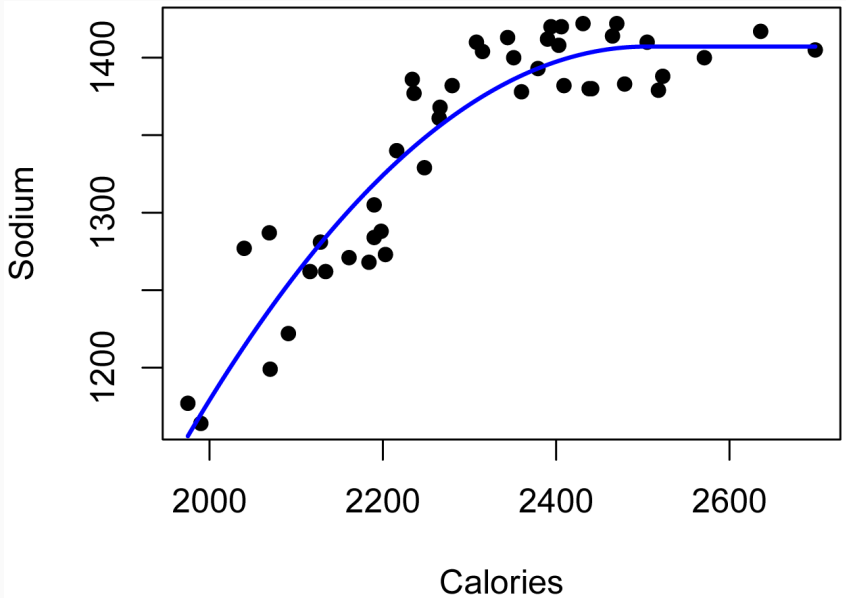
5-fold cross validation

Regression

: estimating the relationships among variables



Linear Regression



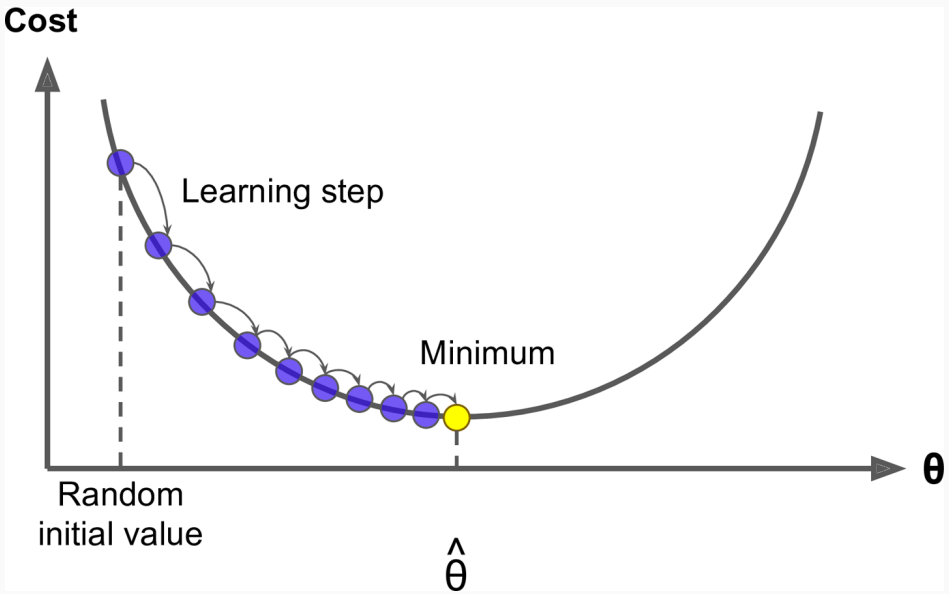
Nonlinear Regression

Gradient Descent

: iterative technique to find a minimum point of the cost function(C).

$$\theta^{(n+1)} = \theta^{(n)} - \alpha_n \nabla C(\theta^{(n)}),$$

where $\theta^{(n)} \in \mathbb{R}^N$ is a model parameter and α_n is a learning rate at n-th iteration.



Stochastic gradient descent (SGD)

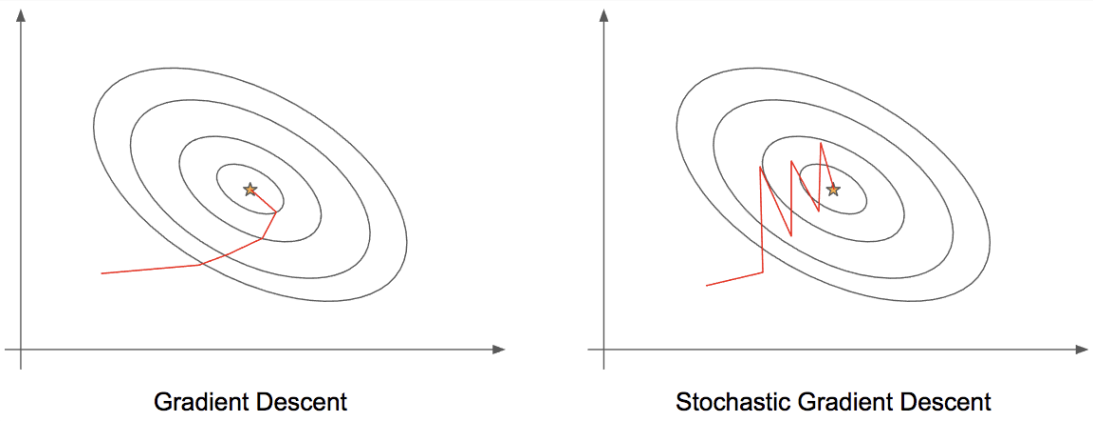
SGD:

$$i(n) \sim \text{Uniform}\{1, \cdots, N\}$$
$$\theta^{(n+1)} = \theta^{(n)} - \alpha_n \nabla C_{i(n)}(\theta^{(n)}),$$

where $C = \sum_{i=1}^{N} C_i$.
$\nabla C_{i(n)}(\theta^{(n)})$ is a stochastic gradient of C at $\theta^{(n)}$, i.e.,

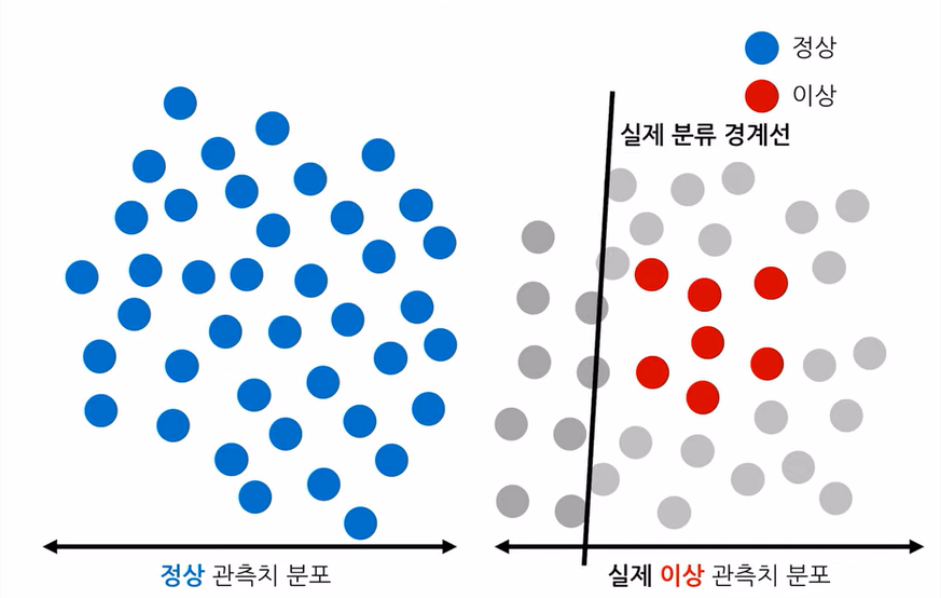
$$\mathbb{E}\big[\nabla C_{i(n)}(\theta^{(n)})\big] = \nabla \mathbb{E}\big[C_{i(n)}(\theta^{(n)})\big] = \nabla C(\theta^{(n)}).$$

When size of the data N is large, SGD is often more effective than GD.



Classification

- Binary classifiers
- Multi-class classifiers
- Multi-label classifiers
- Imbalanced classifiers



Confusion Matrix

TP : TruePositive, FP : FalsePositive,
TN : TrueNegative, FP : FalseNegative

	Classified as **YES**	Classified as **NO**
True **YES**	No. of TPs	No. of FNs
True **NO**	No. of FPs	No. of TNs

$$\text{accuracy} = \frac{\text{number of correctly classified cases}}{\text{total number of cases}} = \frac{TPs + TNs}{\text{total number of cases}}$$

$$\text{precision(정밀도)} = \frac{TPs}{TPs + FPs}$$

$$\text{recall(재현율)} = \frac{TPs}{TPs + FNs}$$

f-score(f-measure) : the harmonic mean of *precision* and *recall*.

Supervised Learning

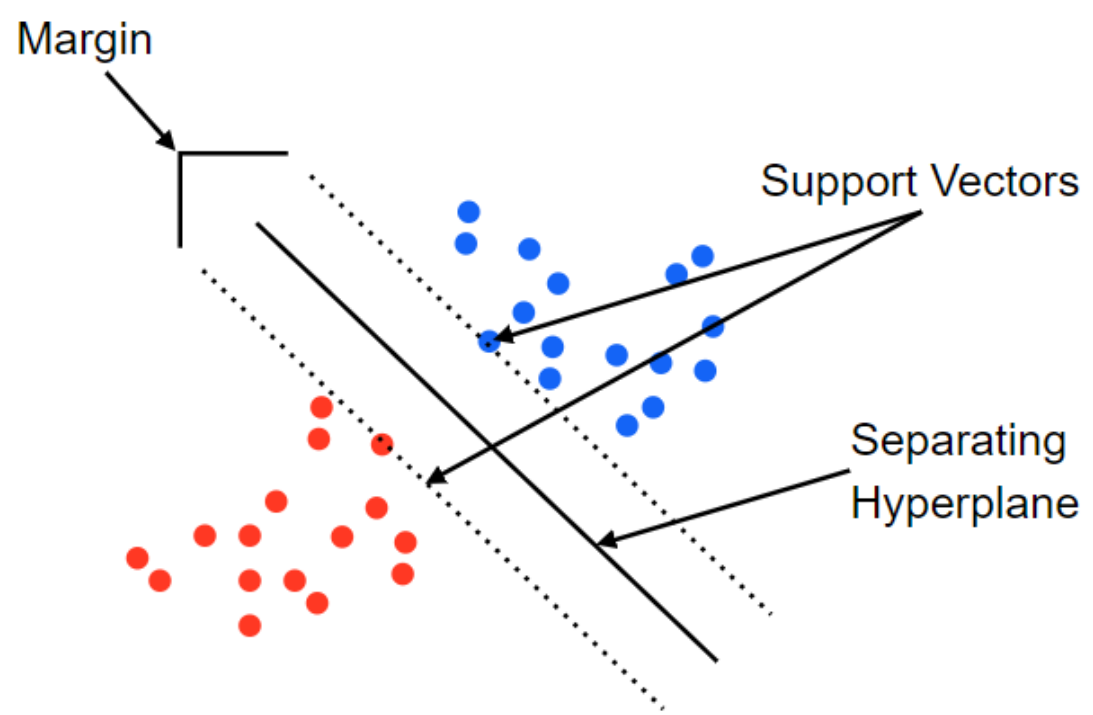
: the machine learning task of learning a function that maps an input to an output based labeled data.

Supervised Learning Methods	Example	Algorithms
Regression	Stock price prediction, House price prediction	Linear regression, ridge, lasso, polynomial
Classification	Spam: Fraud/Not Fraud, Cancer/Not Cancer	Logistic regression, SVM, KNN
Decision Tress	Predict stock prices (regression), classification (credit card fraud)	Decision trees, random forest

Examples of Supervised Learning Algorithms

Support-Vector Machines (SVM)

: constructs a (set of) hyperplanes in a high-dimensional space, which can be used for classification, regression, so on.



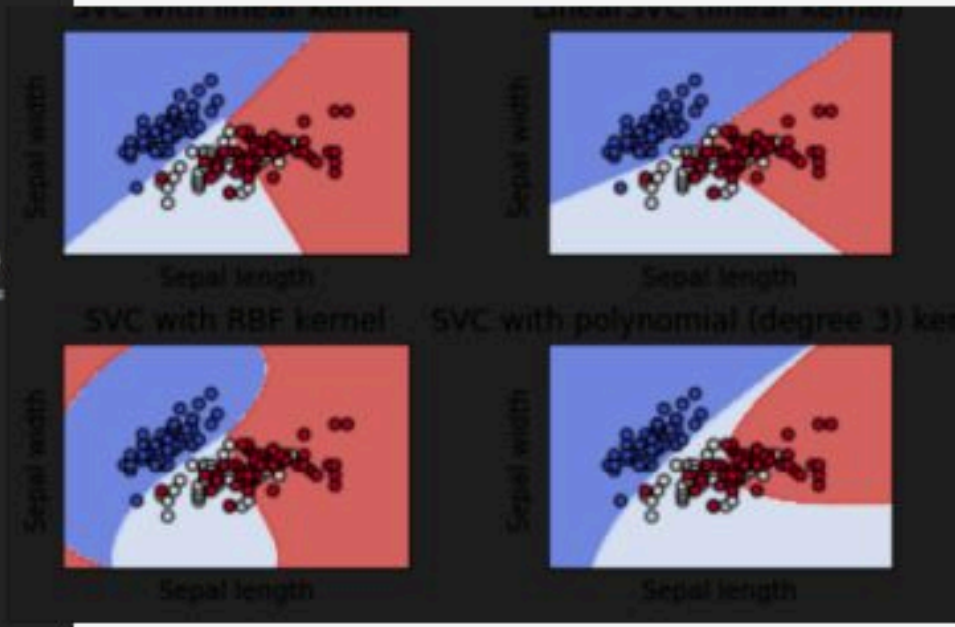
Example(Python): Support-Vector Machines(SVM)

```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn import svm, datasets


def make_meshgrid(x, y, h=0.02):
    """Create a mesh of points to plot in

    Parameters
    ----------
    x: data to base x-axis meshgrid on
    y: data to base y-axis meshgrid on
    h: stepsize for meshgrid, optional

    Returns
    -------
    xx, yy : ndarray
    """
    x_min, x_max = x.min() - 1, x.max() + 1
    y_min, y_max = y.min() - 1, y.max() + 1
    xx, yy = np.meshgrid(np.arange(x_min, x_max, h), np.arange(y_min, y_max, h))
    return xx, yy


def plot_contours(ax, clf, xx, yy, **params):
    """Plot the decision boundaries for a classifier.

    Parameters
    ----------
    ax: matplotlib axes object
    clf: a classifier
    xx: meshgrid ndarray
    yy: meshgrid ndarray
    params: dictionary of params to pass to contourf, optional
    """
    Z = clf.predict(np.c_[xx.ravel(), yy.ravel()])
    Z = Z.reshape(xx.shape)
    out = ax.contourf(xx, yy, Z, **params)
    return out


# import some data to play with
iris = datasets.load_iris()
# Take the first two features. We could avoid this by using a two-dim data
X = iris.data[:, :2]
y = iris.target

# we create an instance of SVM and fit out data. We do not scale our
# data since we want to plot the support vectors
C = 1.0  # SVM regularization parameter
models = (
    svm.SVC(kernel="linear", C=C),
    svm.LinearSVC(C=C, max_iter=10000),
    svm.SVC(kernel="rbf", gamma=0.7, C=C),
    svm.SVC(kernel="poly", degree=3, gamma="auto", C=C),
)
models = (clf.fit(X, y) for clf in models)

# title for the plots
titles = (
    "SVC with linear kernel",
    "LinearSVC (linear kernel)",
    "SVC with RBF kernel",
    "SVC with polynomial (degree 3) kernel",
)
```

Example(Python): Support-Vector Machines(SVM)

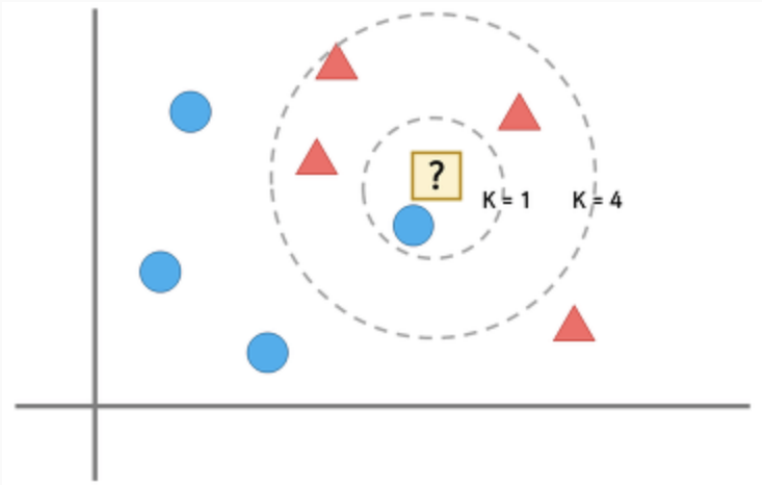
```
# Set-up 2x2 grid for plotting.
fig, sub = plt.subplots(2, 2)
plt.subplots_adjust(wspace=0.4, hspace=0.4)

X0, X1 = X[:, 0], X[:, 1]
xx, yy = make_meshgrid(X0, X1)

for clf, title, ax in zip(models, titles, sub.flatten()):
    plot_contours(ax, clf, xx, yy, cmap=plt.cm.coolwarm, alpha=0.8)
    ax.scatter(X0, X1, c=y, cmap=plt.cm.coolwarm, s=20, edgecolors=
    ax.set_xlim(xx.min(), xx.max())
    ax.set_ylim(yy.min(), yy.max())
    ax.set_xlabel("Sepal length")
    ax.set_ylabel("Sepal width")
    ax.set_xticks(())
    ax.set_yticks(())
    ax.set_title(title)

plt.show()
```

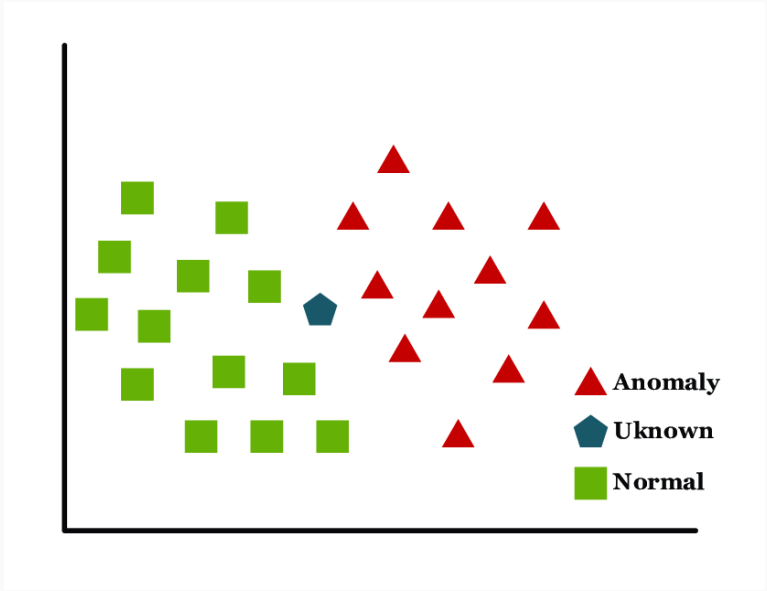
✓ 0.8s



k-Nearest Neighbors (KNN)

: classify the data depending on the nearest k labeled data.





pros) easy and fast

cons) a single error of data can occur poor classification error.

Error and Loss function

x : actual value, $\hat{x}$: predicted values.

$$E_{SSE} = \frac{1}{2}\sum_n \left(x^{(n)} - \hat{x}^{(n)}\right)^2, \quad \text{sum of squared errors}$$

$$E_{MSE} = \frac{1}{n}\sum_n \left(x^{(n)} - \hat{x}^{(n)}\right)^2, \quad \text{mean squared error}$$

$$E_{MAE} = \frac{1}{n}\sum_n \left|x^{(n)} - \hat{x}^{(n)}\right|, \quad \text{mean absolute error}$$

$$E_{BCE} = -\frac{1}{n}\sum_n \left[x^{(n)} \log \hat{x}^{(n)} + \left(1 - x^{(n)}\right) \log \left(1 - \hat{x}^{(n)}\right)\right],$$

Binary cross entropy

Unsupervised Learning

Goal : learn the structure of the dataset which is unlabeled.

Unsupervised Learning Methods	Example	Algorithms
Clustering	Cluster virus data, visitor groups at a conference	Hierarchical clustering, k-means clustering
Dimensionality reduction	Reduction of dimensions in data	Principal component analysis (PCA)

Examples of Unupervised Learning Algorithms

k-Means Clustering

Given an initial set of k means $m_1^{(1)}, \cdots, m_k^{(1)}$,

1. **(Assignment step)**

Assign each observation to the cluster with the nearest mean, i.e.,

$$S_i^{(t)} = \{x_p : \|x_p - m_i^{(t)}\|^2 \leq \|x_p - m_j^{(t)}\|^2, \ \forall j = 1, \cdots, k\},$$

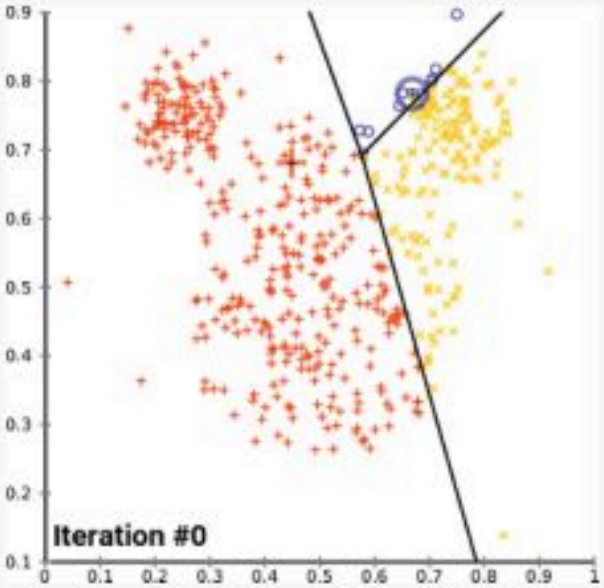
where each x_p is assigned to exactly one $S^{(t)}$.

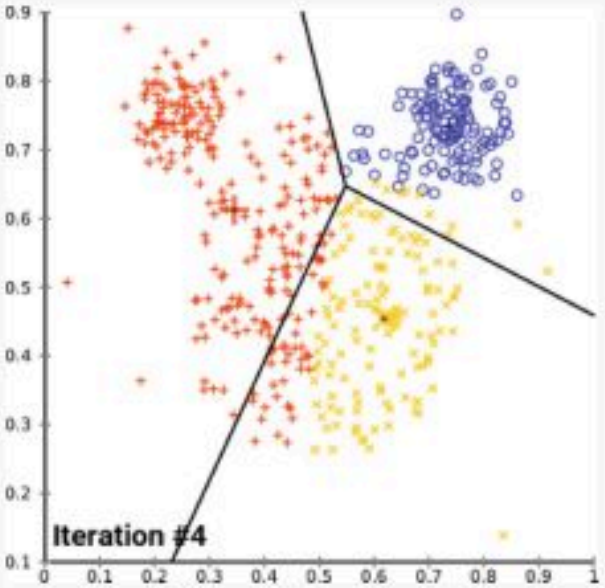
2. **(Update step)**

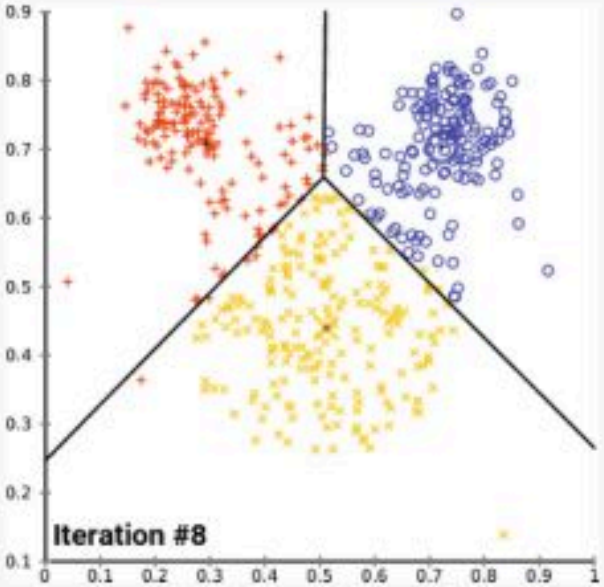
Recalculate means by observations assigned to each cluster.

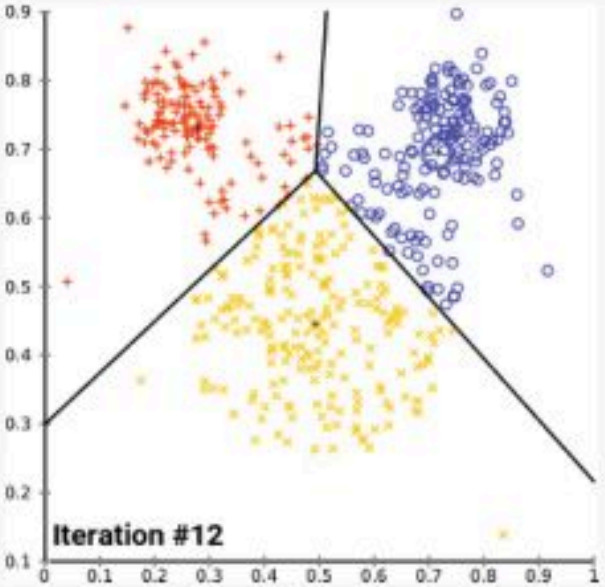
$$m_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{x_j \in S_i^{(t)}} x_j.$$

Example: k-Means Clustering



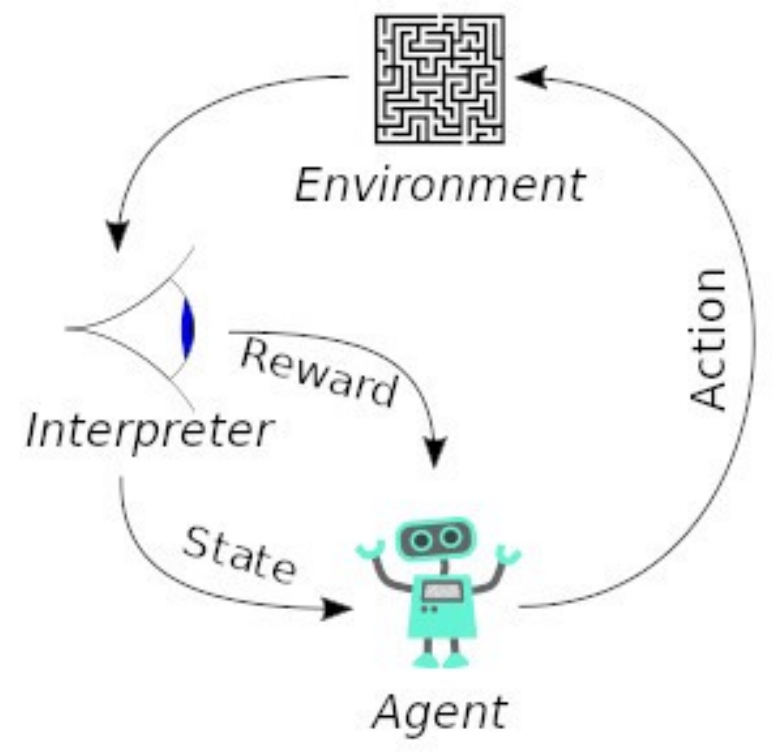






Reinforcement Learning

: Agents are trained on a reward and punishment mechanism.



e.g.) Self-driving cars, personalised product recommendation system, etc.